

Kernel Class

The Kernel of Semantic Kernel.

This is the main entry point for Semantic Kernel. It provides the ability to run functions and manage filters, plugins, and AI services.

Initialize a new instance of the Kernel class.

Constructor

Python

```
Kernel(plugins: KernelPlugin | dict[str, KernelPlugin] | list[KernelPlugin] | None = None, services: AI_SERVICE_CLIENT_TYPE | list[AI_SERVICE_CLIENT_TYPE] | dict[str, AI_SERVICE_CLIENT_TYPE] | None = None, ai_service_selector: AIServiceSelector | None = None, *, retry_mechanism: RetryMechanismBase = None, function_invocation_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE], Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]] = None, prompt_rendering_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE], Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]] = None, auto_function_invocation_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE], Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]] = None)
```

Parameters

 Expand table

Name	Description
plugins	The plugins to be used by the kernel, will be rewritten to a dict with plugin name as key Default value: None
services	The services to be used by the kernel, will be rewritten to a dict with service_id as key Default value: None
ai_service_selector	The AI service selector to be used by the kernel, default is based on order of execution settings. Default value: None
**kwargs Required*	Additional fields to be passed to the Kernel model, these are limited to filters.

Keyword-Only Parameters

 Expand table

Name	Description
<code>retry_mechanism</code> Required*	
<code>function_invocation_filters</code> Required*	
<code>prompt_rendering_filters</code> Required*	
<code>auto_function_invocation_filters</code> Required*	

Methods

 Expand table

<code>add_embedding_to_object</code>	Gather all fields to embed, batch the embedding generation and store.
<code>invoke</code>	Execute a function and return the FunctionResult.
<code>invoke_function_call</code>	Processes the provided FunctionCallContent and updates the chat history.
<code>invoke_prompt</code>	Invoke a function from the provided prompt.
<code>invoke_prompt_stream</code>	Invoke a function from the provided prompt and stream the results.
<code>invoke_stream</code>	Execute one or more stream functions. This will execute the functions in the order they are provided, if a list of functions is provided. When multiple functions are provided only the last one is streamed, the rest is executed as a pipeline.

add_embedding_to_object

Gather all fields to embed, batch the embedding generation and store.

Python

```
async add_embedding_to_object(inputs: TDataModel | Sequence[TDataModel],
    field_to_embed: str, field_to_store: str, execution_settings: dict[str,
```

```
PromptExecutionSettings], container_mode: bool = False, cast_function: Callable[[list[float]], Any] | None = None, **kwargs: Any)
```

Parameters

 Expand table

Name	Description
inputs Required*	
field_to_embed Required*	
field_to_store Required*	
execution_settings Required*	
container_mode	Default value: False
cast_function	Default value: None

invoke

Execute a function and return the FunctionResult.

Python

```
async invoke(function: KernelFunction | None = None, arguments: KernelArguments | None = None, function_name: str | None = None, plugin_name: str | None = None, metadata: dict[str, Any] = {}, **kwargs: Any) -> FunctionResult | None
```

Parameters

 Expand table

Name	Description
function	<xref:semantic_kernel.KernelFunction> The function or functions to execute, this value has precedence when supplying both this and using function_name and plugin_name, if this is none, function_name and plugin_name are used and cannot be None. Default value: None

Name	Description
arguments	<xref:semantic_kernel.KernelArguments> The arguments to pass to the function(s), optional Default value: None
function_name	<xref:<xref:semantic_kernel.str None> > The name of the function to execute Default value: None
plugin_name	<xref:<xref:semantic_kernel.str None> > The name of the plugin to execute Default value: None
metadata	dict[str,<xref: Any>] The metadata to pass to the function(s) Default value: {}
kwargs Required*	dict[str,<xref: Any>] arguments that can be used instead of supplying KernelArguments

Exceptions

 Expand table

Type	Description
KernelInvokeException	If an error occurs during function invocation

invoke_function_call

Processes the provided FunctionCallContent and updates the chat history.

Python

```
async invoke_function_call(function_call: FunctionCallContent, chat_history: ChatHistory, *, arguments: KernelArguments | None = None, execution_settings: PromptExecutionSettings | None = None, function_call_count: int | None = None, request_index: int | None = None, is_streaming: bool = False, function_behavior: FunctionChoiceBehavior = None) -> AutoFunctionInvocationContext | None
```

Parameters

 Expand table

Name	Description
function_call Required*	
chat_history Required*	

Keyword-Only Parameters

 Expand table

Name	Description
arguments Required*	
execution_settings Required*	
function_call_count Required*	
request_index Required*	
is_streaming Required*	
function_behavior Required*	

invoke_prompt

Invoke a function from the provided prompt.

Python

```
async invoke_prompt(prompt: str, function_name: str | None = None, plugin_name: str | None = None, arguments: KernelArguments | None = None, template_format: Literal['semantic-kernel', 'handlebars', 'jinja2'] = 'semantic-kernel', **kwargs: Any) -> FunctionResult | None
```

Parameters

 Expand table

Name	Description
prompt Required*	str The prompt to use
function_name	str The name of the function, optional Default value: None
plugin_name	str The name of the plugin, optional Default value: None
arguments	<xref:<xref:semantic_kernel.KernelArguments None>> The arguments to pass to the function(s), optional Default value: None
template_format	<xref:<xref:semantic_kernel.str None>> The format of the prompt template Default value: semantic-kernel
kwargs Required*	dict[str, <xref: Any>] arguments that can be used instead of supplying KernelArguments

Returns

[Expand table](#)

Type	Description
FunctionResult list[FunctionResult] None	The result of the function(s)

invoke_prompt_stream

Invoke a function from the provided prompt and stream the results.

Python

```
async invoke_prompt_stream(prompt: str, function_name: str | None = None,
plugin_name: str | None = None, arguments: KernelArguments | None = None,
template_format: Literal['semantic-kernel', 'handlebars', 'jinja2'] = 'semantic-
kernel', return_function_results: bool | None = False, **kwargs: Any) ->
AsyncIterable[list[StreamingContentMixin] | FunctionResult | list[FunctionResult]]
```

Parameters

[Expand table](#)

Name	Description
prompt Required*	str The prompt to use
function_name	str The name of the function, optional Default value: None
plugin_name	str The name of the plugin, optional Default value: None
arguments	<xref:<xref:semantic_kernel.KernelArguments None> > The arguments to pass to the function(s), optional Default value: None
template_format	<xref:<xref:semantic_kernel.str None> > The format of the prompt template Default value: semantic-kernel
return_function_results	bool If True, the function results are yielded as a list[FunctionResult] Default value: False
kwargs Required*	dict[str, <xref: Any>] arguments that can be used instead of supplying KernelArguments

Returns

[Expand table](#)

Type	Description
AsyncIterable[StreamingContentMixin]	The content of the stream of the last function provided.

invoke_stream

Execute one or more stream functions.

This will execute the functions in the order they are provided, if a list of functions is provided. When multiple functions are provided only the last one is streamed, the rest is executed as a pipeline.

Python

```

async invoke_stream(function: KernelFunction | None = None, arguments:
KernelArguments | None = None, function_name: str | None = None, plugin_name: str
| None = None, metadata: dict[str, Any] = {}, return_function_results: bool =
False, **kwargs: Any) -> AsyncGenerator[list[StreamingContentMixin] |
FunctionResult | list[FunctionResult], Any]

```

Parameters

[Expand table](#)

Name	Description
function	<xref:semantic_kernel.KernelFunction> The function to execute, this value has precedence when supplying both this and using function_name and plugin_name, if this is none, function_name and plugin_name are used and cannot be None. Default value: None
arguments	<xref:<xref:semantic_kernel.KernelArguments None>> The arguments to pass to the function(s), optional Default value: None
function_name	<xref:<xref:semantic_kernel.str None>> The name of the function to execute Default value: None
plugin_name	<xref:<xref:semantic_kernel.str None>> The name of the plugin to execute Default value: None
metadata	dict[str,<xref: Any>] The metadata to pass to the function(s) Default value: {}
return_function_results	bool If True, the function results are yielded as a list[FunctionResult] Default value: False
content Required*	content (<xref:in addition to the streaming>)
yielded. Required*	yielded. (<xref:otherwise only the streaming content is>)
kwargs Required*	dict[str,<xref: Any>] arguments that can be used instead of supplying KernelArguments

Attributes

retry_mechanism

Data descriptor used to emit a runtime deprecation warning before accessing a deprecated field.

Python

```
retry_mechanism: RetryMechanismBase
```

function_invocation_filters

Filters applied during function invocation, from KernelFilterExtension.

Python

```
function_invocation_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE, Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]]]
```

prompt_rendering_filters

Filters applied during prompt rendering, from KernelFilterExtension.

Python

```
prompt_rendering_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE, Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]]]
```

auto_function_invocation_filters

Filters applied during auto function invocation, from KernelFilterExtension.

Python

```
auto_function_invocation_filters: list[tuple[int, Callable[[FILTER_CONTEXT_TYPE, Callable[[FILTER_CONTEXT_TYPE], Awaitable[None]]], Awaitable[None]]]]
```

plugins

A dict with the plugins registered with the Kernel, from KernelFunctionExtension.

services

A dict with the services registered with the Kernel, from KernelServicesExtension.

ai_service_selector

The AI service selector to be used by the kernel, from KernelServicesExtension.

msg

The deprecation message to be emitted.

wrapped_property

The property instance if the deprecated field is a computed field, or *None*.

field_name

The name of the field being deprecated.